



CS & IT ENGINEERING



Algorithms

Analysis of Algorithms

Lecture No.- 04



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Asymptotic Notations
→ O, Ω, Θ

Topics to be Covered



Topic

Topic

Topic

Practice

Mathematical Properties

Small Notations



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.

Telegram



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

TTTe

17 Big-Oh (O)

$$f(n) = O(g(n))$$

iff,

$$f(n) \leq C * g(n)$$

some $C > 0$



Topic : Asymptotic Notations

(2) Big Omega (Ω):

$f(n)$ & $g(n)$

$$f(n) = \Omega(g(n))$$

iff, $f(n) \geq c * g(n)$; $n > n_0$, $c > 0$

e.g.: $7n^3 + 8n + 1000 \Rightarrow \Omega(n^3)$



Topic : Asymptotic Notations

(3) Theta (θ):

Tight Bound

$f(n) \& g(n) \Rightarrow f(n) = \theta(g(n))$

iff, $f(n) = O(g(n))$

and $f(n) = \Omega(g(n))$

or $c_2 * g(n) \leq f(n) \leq c_1 * g(n)$

Hence, $f(n) = \theta(g(n))$

e.g.: $9n^3 + 7n^2 + 100$

$\Rightarrow f(n) = O(n^3)$

$= \Omega(n^3)$



Topic : Exponentials



For all real $a > 0$, m , n

$$a^0 = 1$$

$$a^1 = a$$

$$a^{-1} = 1/a$$

$$(a^m)^n = a^{mn}$$

$$(a^m)^n = (a^n)^m$$

$$a^m \cdot a^n = a^{m+n}$$

$$a^{-2} = \frac{1}{a^2}$$



Topic : Exponentials



$$(a^m)^n = a^{m \times n}$$

e.g.:

$(2^3)^2$	$= 2^{3 \times 2}$	$= 2^{2 \times 3}$	$= (2^2)^3$
$(8)^2$	$= 2^6$	$= 2^6$	$= 4^3$
(64)	$= 64$	$= 64$	$= 64$



Topic : Exponentials

$$a^m \times a^n = a^{(m+n)} \neq a^{\underline{(m \times n)}}$$

e.g.: $2^3 \times 2^2 = 2^{(3+2)}$

$$8 \times 4 = 2^5$$

$$32$$

$$= 32$$



Topic : Analysis of Algorithms



Logarithmic Properties:

$$\log X^y = y \log x$$

$$\log(xy) = \log x + \log y$$

$$\log \log n = \log (\log n)$$

$$a^{\log_b x} = x^{\log_b a}$$

$$a = b^{\log_b a} = b^{\log_b a} = a^{\log_b b} = a$$

$$\log_b a^n = n \cdot \log_b a$$

$$\log_b a = \frac{1}{\log_a b}$$

$$\log n = \log_{10} n$$

$$\log^k n = (\log)^k n$$

$$\log\left(\frac{x}{y}\right) = \log x - \log y$$

$$\log_b^x = \frac{\log_a^x}{\log_a^b}$$

$$\log_c(ab) = \log_c a + \log_c b$$

$$\log_b^{(1/a)} = -\log_b a$$

$$a^{\log_b c} = c^{\log_b a}$$



Topic : Asymptotic Notations

Example: $\log_2(16)$

$$\Rightarrow \log_2(4 \times 4) = \log_2 4 + \log_2 4 = 2 + 2 = \underline{4}$$

$$\Rightarrow \log_2(2^4) = \log_2 2^2 + \log_2 2^2 = 4$$

$$= 4 \times \log_2(2)$$

4



Topic : Asymptotic Notations

Logarithmic Properties:

$$\log\left(\frac{a}{b}\right) = \log(a) - \log(b)$$

$$\log(a * b) = \log(a) + \log(b)$$

$$\log_b(a) = \frac{\log_c(a)}{\log_c(b)} = \frac{1}{\frac{\log_c(b)}{\log_c(a)}} = \frac{1}{\log_a(b)}$$

$$\frac{3}{2} = \frac{1}{2/3}$$



Topic : Asymptotic Notations

Logarithmic Properties:

$$\log^2(n) = [\log(n)]^2 = \underbrace{(\log n) * (\log n)}$$

$$\log_b \left(\frac{1}{a} \right) = \log_b(1) - \log_b(a)$$

$$= 0 - \log_b(a)$$

$$= -\log_b a$$

$$\log^2(n) = \log(\log n) \quad \times$$

Geometric Progression



Topic : Geometric Sum Formula

1. The geometric sum formula for finite terms is given as:

if $r = 1$, $S_n = n \cdot a$

if $|r| < 1$, $S_n = \frac{a(1-r^n)}{1-r}$

if $|r| > 1$, $S_n = \frac{a(r^n-1)}{r-1}$

Where

- a is the first term
- r is the common ratio
- n is the number of terms

$$\underline{2, 2, 2, 2}$$

$$2 \times 4$$



Topic : Asymptotic Notations

Geometric Progression:

$$S = 2^1 + 2^2 + 2^3 + 2^4 = 2 + 4 + 8 + 16 = 10 + 20 = 30$$

Or, $r = 2$ and $n = 4$

$a = 2$

$$\text{Sum} = \frac{a(r^n - 1)}{(r - 1)} = \frac{2(2^4 - 1)}{(2 - 1)} = \frac{2(16 - 1)}{1} = 30 \checkmark$$



Topic : Asymptotic Notations

Example:

$$\sum_{i=1}^n 2^i = 2^1 + 2^2 + \dots + 2^n$$

n terms, $a = 2^1 = 2$, $r = 2$

$$S_n = \frac{a(r^n - 1)}{(r - 1)} = \frac{2(2^n - 1)}{2 - 1} = 2(2^n - 1) \checkmark$$



Topic : Asymptotic Notations

Example:

$$\sum_{i=1}^n \frac{1}{3^i} = \frac{1}{3^1} + \frac{1}{3^2} + \dots + \frac{1}{3^n}$$

n terms, $a = 1/3^1 = 1/3$, $r = 1/3$ */ n = n*

$$S_n = \frac{a(1 - r^n)}{(1 - r)} = \frac{\frac{1}{3} \left(1 - \left(\frac{1}{3} \right)^n \right)}{1 - \frac{1}{3}} = \frac{1}{2} \left(1 - \frac{1}{3^n} \right)$$



Topic : Geometric Sum Formula

2. The geometric sum formula of infinite terms is given as:

if $|r| < 1$ $S_{\infty} = \frac{a}{1-r}$

if $|r| > 1$, the series does not converge and it has no sum.



Topic : Analysis of Algorithms

Arithmetic series

V. Imp

$$\sum_{k=1}^n k = 1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

Geometric series

$$\sum_{k=0}^n x^k = 1 + x + x^2 + \dots + x^n = \frac{x^{n+1} - 1}{x - 1} \quad (x \neq 1)$$

Harmonic series

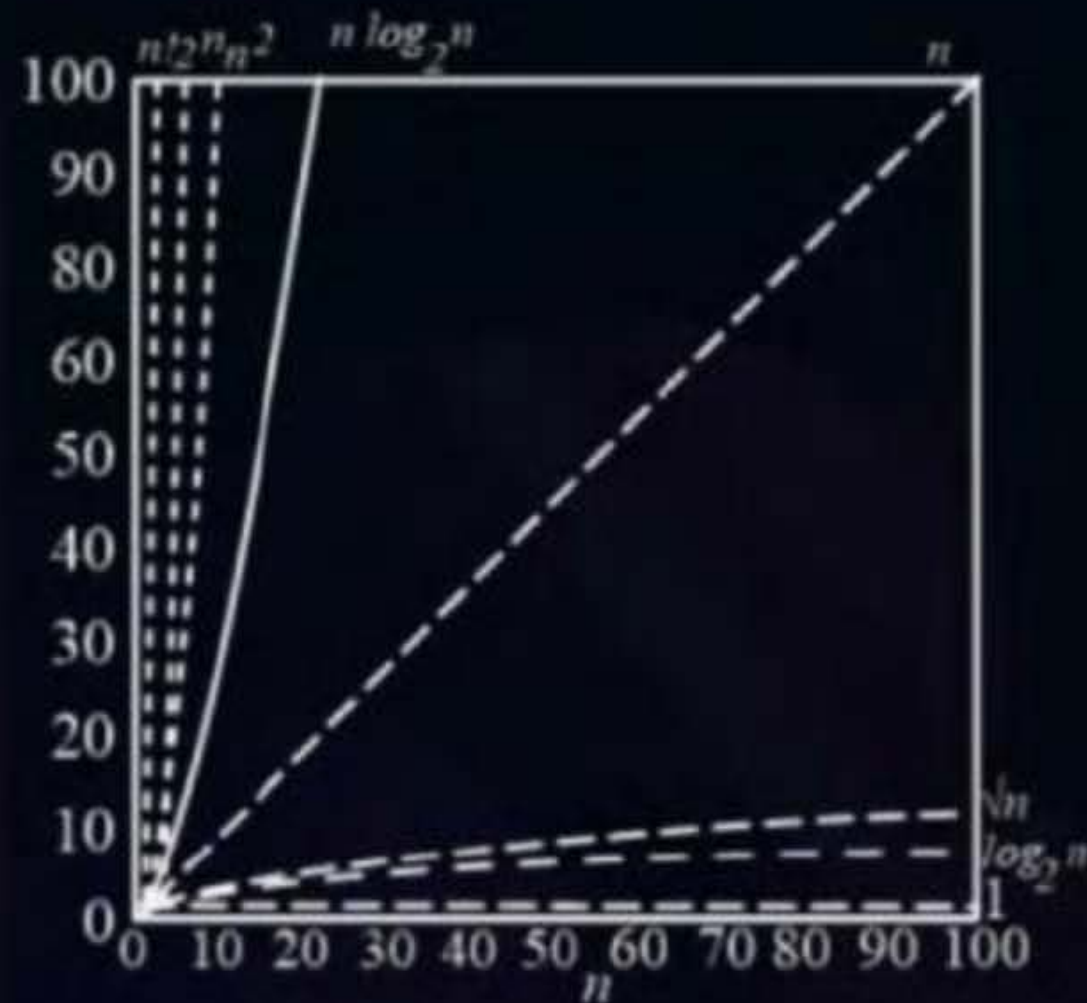
$$\sum_{k=1}^n \frac{1}{k} = \left(1 + \frac{1}{2} + \dots + \frac{1}{n} \right) \approx \log n$$



Topic : Analysis of Algorithms

Dominance Relation

Constant < logarithms < poly < exponential



Practice



Topic : Analysis of Algorithms

Practice Questions:

$$(1) f(n) = \sum_{a=1}^n a$$

↓
0
2
0

$$f(n) = \sum_{a=1}^n a = 1 + 2 + 3 + 4 + \dots + n$$

$$= \frac{n(n+1)}{2} = \frac{n^2 + n}{2} = O(n^2)$$

$\sqrt{2}(n^2) > \underline{O(n^2)}$



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Practice Questions:

$$(2) \underline{f(n)} = \sum_{a=1}^n a^2$$

$$f(n) = 1^2 + 2^2 + 3^2 + \dots + n^2$$

$$\begin{aligned} &= \frac{n(n+1)(2n+1)}{6} = \frac{(n^2+n)(2n+1)}{6} = \frac{2n^3 + n^2 + 2n^2 + n}{6} \\ &= \frac{2n^3 + 3n^2 + n}{6} \end{aligned}$$

$\Theta(n^3)$ (indicated by an arrow pointing to the n^3 term in the numerator of the final expression)

$= O(n^3) > \underline{\Omega(n^3)}$ (indicated by an arrow pointing from the final expression to this line)



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Practice Questions:

$$(3) f(n) = \sum_{a=1}^n a^3$$

$$= 1^3 + 2^3 + 3^3 + \dots + n^3$$

$$= \left[\frac{n(n+1)}{2} \right]^2$$

$$O(n^4)$$

$$O(n^4) \sim \Omega(n^4)$$

$$= \frac{n^2(n+1)^2}{4} = \frac{n^2(n^2 + 2n + 1)}{4}$$

$$= \frac{n^4 + 2n^3 + n^2}{4} \Rightarrow \underline{O(n^4)}$$



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Practice Questions:

$$(4) f(n) = \sum_{a=1}^n 1$$

$$= \underbrace{1 + 1 + \dots + 1}_{n \text{ times}}$$

$$= n \times 1$$

$$= n$$

$$\Theta(n) \begin{cases} \rightarrow O(n) \\ \rightarrow \Omega(n) \end{cases}$$



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Practice Questions:

$$(5) f(n) = \sum_{a=1}^n 3^a$$

$$= 3^1 + 3^2 + \dots + 3^n$$

$$\left. \begin{array}{l} a=3 \\ r=3 \\ h=n \end{array} \right\}$$

$$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{3(3^n - 1)}{3 - 1} = \frac{3(3^n - 1)}{2} = O(3^n)$$

$$\begin{aligned} 3^{n+1} &= \boxed{\Theta(3^{n+1})} \\ &= \Theta(3^n) \end{aligned}$$

$$\frac{O(3^n)}{2} \rightarrow \Omega(3^n)$$

⑥ $F(n) = \sum_{i=1}^n \left(\frac{1}{5}\right)^i$
(v.v.imp)

$O\left(\frac{1}{5}n\right) \times$

$$= \frac{1}{5} + \frac{1}{5^2} + \frac{1}{5^3} + \dots + \frac{1}{5^n}$$

$a = \frac{1}{5}$
 $r = \frac{1}{5}$
 $n = n$

$|r| < 1$

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{\frac{1}{5} \left(1 - \frac{1}{5^n}\right)}{1 - \frac{1}{5}} = \boxed{\frac{1}{4} \left(1 - \frac{1}{5^n}\right)}$$

$\frac{1}{5^n} \rightarrow \text{decr}$

$\frac{1}{4} \rightarrow \text{const}$



$O(1)$
 $\swarrow \searrow$
 $O(1)$



Topic : Analysis of Algorithms

Practice Questions:

$$(7) f(n) = \sum_{a=1}^n n$$

$$= n + n + n \dots + n$$

$$= n * n = n^2$$

$$f(n) = n^2 \rightarrow \Theta(n^2) \begin{cases} \nearrow O(n^2) \\ \searrow \Omega(n^2) \end{cases}$$



Topic : Analysis of Algorithms

1. Big Oh (UB)

$$f(n) = O(g(n))$$

Iff $f(n) \leq c * g(n)$, some $c > 0$, $n \geq n_0$

2. Big Omega (Ω)

$$f(n) = \Omega(g(n))$$

Iff $f(n) \geq c * g(n)$, some $c > 0$, $n \geq n_0$

3. $F(n) = \Theta(g(n))$

if $f(n) = O(g(n))$

and

$$F(n) = \Omega(g(n))$$



Topic : Analysis of Algorithms

1. $f(n) = \prod_{i=1}^n (1) = \underbrace{1 \times 1 \times 1 \times \dots \times 1}_{n \text{ times}} = 1$

$\Pi \rightarrow$ Product

Summation

$(\text{sum}) \sum_{i=1}^n 1 \Rightarrow \underbrace{1 + 1 + \dots + 1}_{n \text{ times}}$

$\Rightarrow n$

$= O(n)$

$\Rightarrow 0(1) \rightarrow$ Constant TC



Topic : Analysis of Algorithms

#Q. $f(n) = \prod_{i=1}^n (i)$

46%

$$= 1 \times 2 \times 3 \cdots n$$

$$= n!$$

$$= \underline{\underline{O(n^n)}}$$

A

$$F(n) = O(n)$$

B

$$F(n) = O(n^2)$$

C

$$F(n) = O(n^3)$$

☒ **D**

$$F(n) = O(n^n)$$



Topic : Analysis of Algorithms

$$f(n) = \prod_{i=1}^n (i) = 1 \times 2 \times 3 \times 4 \dots n$$

$$= \boxed{n!}$$

$$F(n) = O(n!)$$

$$= O(n^n)$$



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$n!$ vs n^n

$$n \leq n, n \geq 2$$

$$n(n-1) \leq n \times n$$

$$n * (n-1) * (n-2) \leq n * n * n$$

$$n * (n-1) * \dots * 1 \leq n * n * n * \dots * n \text{ (n times)}$$

$$n! \leq (n^n)$$

$$n! = O(n^n)$$

$$\text{Is } n! = \Omega(n^n)?$$



Topic : Analysis of Algorithms

#Q. Is $n! = \Omega(n^n)$?

A

~~True $\rightarrow$ 30%~~

B

~~False $\rightarrow$ 66%~~

Imp. Note

$$n! = O(n^n)$$

$$\text{and } n! \neq \Omega(n^n)$$

$$\text{Hence, } n! \neq \Omega(n^n)$$

$$n * (n-1) * (n-2) * \dots * 1 \geq c * n^n ?$$

Never

$$f(n) \geq c * g(n) \rightarrow \times$$

$$n! \neq \Omega(n^n)$$



Topic : Analysis of Algorithms

#Q. $f(n) = \prod_{i=1}^n [\log(i)]$

$$= \log(1) * \log(2) * \dots * \log(n)$$

$$= 0 * \text{anything}$$

$$= 0 \rightarrow \theta(1)$$

$$\begin{array}{l} \theta(1) \leftarrow \Omega(1) \\ \theta(1) \leftarrow O(1) \\ \downarrow \end{array}$$

A $F(n) = O(\log n)$

B $F(n) = \cancel{O(\log n)} O(n \log n)$

C $F(n) = \Omega(\log n)$

D $F(n) = \cancel{\theta(\log n)} \theta(\log n) X$

$$\begin{array}{l} O(1) \\ O(n) \\ \downarrow \end{array}$$



Topic : Analysis of Algorithms

#Q. $f(n) = \sum_{i=1}^n \log(i)$

A

$F(n) = O(\log n)$

B

$F(n) = \cancel{O(\log n)} \quad O(n \log n)$

C

$F(n) = \cancel{\Omega(\log n)} \quad \Omega(n \log n)$

D

$F(n) = \cancel{\Theta(\log n)} \quad \Theta(n \log n)$



Topic : Analysis of Algorithms

Sol. $f(n) = \sum_{i=1}^n [\log(i)]$

$$= \log(1) + \log(2) + \log(3) + \dots + \log(n)$$

$$= \log(1 * 2 * 3 * \dots * n)$$

$$f(n) = \log(n!).$$

$$\approx \log(n^n)$$

$$\approx \boxed{n \log n} \rightarrow O(n \log n)$$

$$\log(a) + \log(b) = \log(a * b)$$



Topic : Analysis of Algorithms

Method:- 1.

Logic

$$\log(n) \leq \log(n)$$

$$\log(n) + \log(n-1) \leq \log(n) + \log(n)$$

$$\log(n) + \log(n-1) + \dots + \log(1) \leq \log(n) + \log(n) + \dots + \log(n) \rightarrow \text{n times}$$

$$\log(n) + \dots + \log(1) \leq n * \log(n)$$

$$\log(n!) \leq n * \log(n) \Rightarrow \log(n!) = O(n \log n)$$



Topic : Analysis of Algorithms

Method:- 2.

Stirling's

Using ~~string's~~ Approximation

⇓

$$n! \approx \sqrt{2\pi n} * \left(\frac{n}{e}\right)^n$$



Topic : Analysis of Algorithms

$$f(n) = \sum_{i=1}^n \log(i) = \log(n) \quad \log(n!)$$

$$= \log \left(\sqrt{2\pi n} * \left(\frac{n}{e} \right)^n \right) \rightarrow \text{using stirting's Approx.}$$

$$= \log(\sqrt{2\pi}) + \frac{1}{2} \log(n) + n[\log(n) - \log(e)]$$

$$= \left[\log(\sqrt{2\pi}) + \frac{1}{2} \log(n) + n * \log(n) - n * \log(e) \right]$$

$$\begin{array}{l} f(n) = O(n \log n) \\ f(n) = \Omega(n \log n) \end{array} \rightarrow \log(n!) = \underline{\underline{\theta(n \log n)}}$$

dominating term



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Bonus logic → Method -3 → Using Integration

$$f(n) = \sum_{i=1}^n \log(i) = \log(1) + \log(2) + \dots + \log(n)$$

$$= \int_1^n \log(x) dx = [x(\log x - 1)]_1^n$$

$$= n \log n - n + c$$

$$= O(\log n)$$

$$= \underline{O(n \log n)}$$



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H.W

(PYQ)

#Q. $f(n) = \sum_{i=1}^n i^3 = x \text{ cohoice for } x.$

I. $\theta(n^4)$

II. $\theta(n^5)$

III. $O(n^5)$

IV. $\Omega(n^3)$

A

I, II, II

B

II, III, IV

C

I, II, III, IV

D

I, III, IV



Topic : Analysis of Algorithms

2. Small notations always provide bounds that are loose bounds.



(never tight bound)



Summary



- Practice
- Maths Properties
- mcqs.



THANK - YOU